

AMPHIBIANS



COLD-BLOODED animals, amphibians are vertebrates (animals with a backbone) that evolved from fish.

They are adapted for life on land, but most must return to water in some form to breed. Amphibians undergo a process known as metamorphosis in their development from larvae to adult, hence the Greek origin of their name: *amphi* meaning "double"; *bios* meaning "life". There are three groups of amphibians and more than 3,000 species.

Couch's spadefoot toad



Distribution of amphibians

Amphibians live everywhere. Desert species survive the driest season by staying underground inside a membranous sac, which they secrete themselves. Some temperate species hibernate in pond mud in the winter.

Amphibian groups

There are three groups of amphibians: the worm-like caecilians; the tailed amphibians, including newts and salamanders; and the tail-less frogs and toads, probably the most diverse group.



Caecilians

Caecilians are legless, carnivorous amphibians most of which live in the tropics. Some species burrow in the ground; others are aquatic. They have small eyes and ears and sensory tentacles on the head.

Skin

Amphibian skin is thin and scaleless. It is usually kept moist with mucus to increase its ability to allow oxygen through for skin breathing. Skin can be smooth or rough. It secretes certain chemicals: pheromones can attract potential mates, while poisons deter predators. As they grow, amphibians shed the top layer of skin.



Great crested newt Square marked toad



Mandarin salamander Tree frog

Texture

Many frogs and toads have smooth skin covered by mucus. Other amphibians, such as the mandarin salamander and many dry-skinned toads, have raised nodules.



White's tree frogs

Colour

Amphibians may have skin colours that absorb or reflect heat. Colour also varies with temperature, becoming pale when warm and darker if cold and damp.

Camouflage

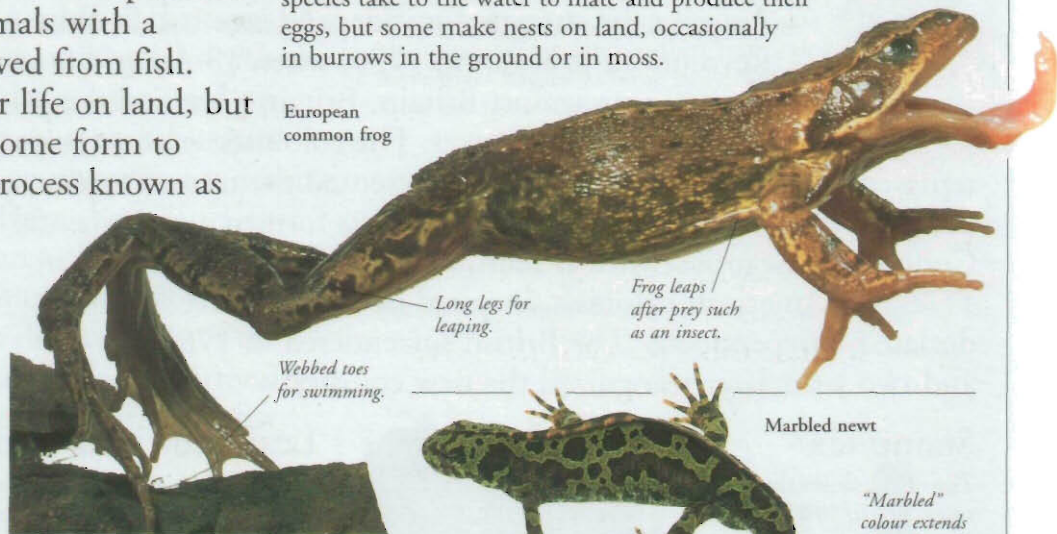
Many frogs and toads are camouflaged to avoid detection by predators. Most have a combination of forest colours and disruptive patterning. Some rainforest species are shaped to look like dead leaves.



Poison-dart tadpoles

Amphibian features

Apart from the caecilians and a few species of salamander, adult amphibians have four legs, each with four or five digits. Most species take to the water to mate and produce their eggs, but some make nests on land, occasionally in burrows in the ground or in moss.



European common frog

Long legs for leaping.

Frog leaps after prey such as an insect.

Webbed toes for swimming.

Marbled newt

"Marbled" colour extends along the tail.



Frogs and toads

In temperate regions, frogs are more aquatic than toads, have slimier skin and longer legs. In the tropics, some species of frog and toad are fully aquatic and live in trees or underground.

Newts and salamanders

The tailed amphibians – newts, salamanders, and the eel-like sirens of North America – live in tropical forests, temperate woods, mountain streams, and lakes. Some have very specialized lifestyles: a few even live in the total darkness of caves.

Metamorphosis

The development from an aquatic larva that breathes through gills, or spiracles, to an air-breathing adult is called metamorphosis. It involves the growth of legs and the loss of the tail in frogs and toads.

Eggs

Amphibian eggs are laid singly, in clumps, or in strings of clear "jelly" called spawn. They have no shell and require a moist environment to survive.

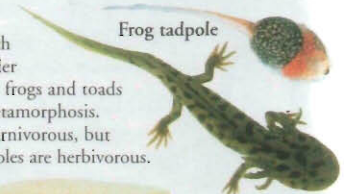
Frog spawn



Tadpoles

Larvae, or tadpoles, hatch from the eggs. Salamander tadpoles have limbs, but frogs and toads develop these during metamorphosis. Salamander larvae are carnivorous, but most frog and toad tadpoles are herbivorous.

Frog tadpole



Gills

Salamander tadpole



Axolotl

Some salamanders may stay as larvae all their life. The axolotl is a form of the Mexican tiger salamander.

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